

9 Facial Plastic and Reconstructive Surgery

Section Editor

Jessyka G. Lighthall

Contributors

Daniel G. Becker

Ara A. Chalian

Donn R. Chatham

John L. Frodel Jr.

David Goldenberg

Bradley J. Goldstein

Robert M. Kellman

Ayesha N. Khalid

Christopher K. Kolstad

Theda C. Kontis

J. David Kriet

Phillip R. Langsdon

Jessyka G. Lighthall

E. Gaylon McCollough

Michael P. Ondik

Stephen S. Park

Francis P. Ruggiero

John M. Schweinfurth

Dhave Setabutr

Scott J. Stephan

Jonathan M. Sykes

Travis T. Tollefson

Robin Unger

Jeremy Watkins

9.1 Craniomaxillofacial Trauma

For ear and temporal bone trauma, see **Chapter 3.1.2**. For laryngeal fractures, see **Chapter 5.1.2**. For neck trauma, see **Chapter 6.1.4**.

9.1.1 Nasal Fractures

◆ Key Features

- Nasal fractures are the most common head and neck fracture.
- They have aesthetic and functional implications.
- Septal hematoma should be recognized early and managed immediately.
- Closed reduction may reduce the need for delayed treatment.
- Open reduction may be necessary.
- Even if no surgical intervention is initially planned, the patient should be reevaluated in 1 to 2 weeks after any traumatic swelling has resolved.
- Cartilaginous injury, persistent deformity, or persistent nasal obstruction may require delayed functional septorhinoplasty.

The clinical presentation of a nasal fracture may include a history of nasal trauma with associated pain, edema, epistaxis, change in external nasal appearance, nasal airway obstruction, and infraorbital ecchymosis. Imaging is not necessary to diagnose a nasal fracture but may be indicated to rule out other injuries. Deformities that are not immediately identified may become obvious as edema resolves. Treatment may be immediate or delayed based on history, exam, and patient desires. Even if immediate intervention is performed, appropriate follow-up is necessary to identify persistent deformities or nasal obstruction.

◆ Epidemiology

Nasal fractures are cited as the most common type of facial fracture, accounting for approximately half of all facial fractures. These injuries occur largely in the younger, physically active segments of the population and predominantly in males. The most common mechanism is blunt trauma (e.g., accidents, assault, sports), although these fractures also occur via penetrating and high-energy injuries. Nasal fractures occur not only in isolation but also frequently in conjunction with more extensive facial fractures.

◆ Evaluation

A detailed history on mechanism of injury, patient symptoms, associated complaints, premorbid status of the nose, and a general health history should be obtained.

Physical examination is critical in the diagnosis of nasal fractures and may be facilitated if the nose is decongested. Visual inspection, manual palpation, and anterior rhinoscopy are essential. Nasal endoscopy may be performed to increase the acquisition of meaningful clinical data when necessary (**Fig. 9.1**).

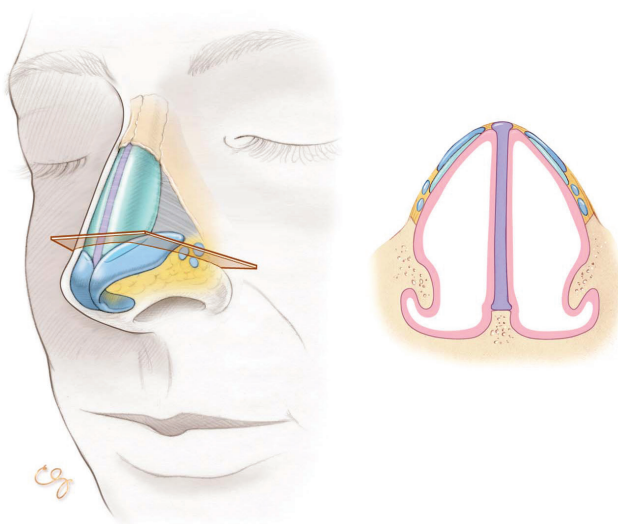


Fig. 9.1 Cross-section of the nose. Note the supporting soft tissue structures and attachment to the lateral nasal wall. Preservation of the lower portion of the lateral bony nasal wall is critical to preventing nasal airway narrowing postoperatively. (Used with permission from Papel ID, ed. *Facial Plastic and Reconstructive Surgery*. 4th ed. New York, NY: Thieme; 2016:439.)

After a focused facial trauma exam to rule out associated injuries, the nose should be examined. The integrity of the nasal skin should be assessed. Often, nasal edema limits visual examination. Palpation of the nose is critical to making the correct diagnosis. All aspects of the nose should be evaluated. Globally, the examiner should determine whether the nose is straight, whether there is a deviation, or whether other dorsal deformities exist (e.g., C-shaped deformity, saddle deformity, etc.). Palpation of the nasal bones will identify fracture of the bony pyramid as a segment versus more complex comminuted injuries. The examiner should assess rotation, projection, and tip and sidewall support. Anterior rhinoscopy must be performed to assess

the status of the septum. Determining the status of the septum is frequently underemphasized; however, appropriate appreciation and management of septal injuries is essential to the restoration of optimal nasal function and appearance. The examiner should look for and document the presence or absence of a septal hematoma. If present, it should be expeditiously and appropriately managed.

Photographs should be taken, similar to the views obtained for rhinoplasty evaluation. Planar radiographs and computed tomography (CT) rarely add more valuable data than those obtained through the physical examination and medical history unless an associated injury is suspected based on exam (e.g., telecanthus, dystopia).

◆ Treatment Options

Observation

If minimal displacement of the nasal bones, minimal soft tissue injury, and minimal compromise of the nasal airway exists, observation is appropriate. Patients should be followed within 1 to 2 weeks once edema has resolved to confirm that no posttraumatic deformity exists.

Closed Manipulation

In the case of obvious bony deviation or more severe nasal bone or septal injuries, a closed reduction, with or without splinting, may be warranted. This may be performed under local or general anesthesia, based on patient and surgeon preference. Manipulation is typically performed within 2 to 3 days of injury prior to bony healing of the fracture. During closed reduction, the patient's fractured nasal bones and septum are mobilized and reduced digitally. This may be aided with the use of blunt instruments such as the Boies elevator or Ashe forceps. Traditional closed manipulation is best applied in patients who have actual subluxation or displacement of the nasal bones without comminution of the nasal bones themselves. However, closed reduction may be attempted in the latter case with the use of intranasal and extranasal stabilization.

Modified Open Reduction with Osteotomies

The modified open technique is a limited version of an open technique in which intranasal incisions are made for the introduction of osteotomes. Such patients frequently benefit from manipulation of the nasal bones into position after undergoing bilateral micro-osteotomies. As with closed manipulation, the position of the septum may impede the success of this technique. A secondary procedure may then be necessary or other treatment offered.

Open Nasal/Septal Repair

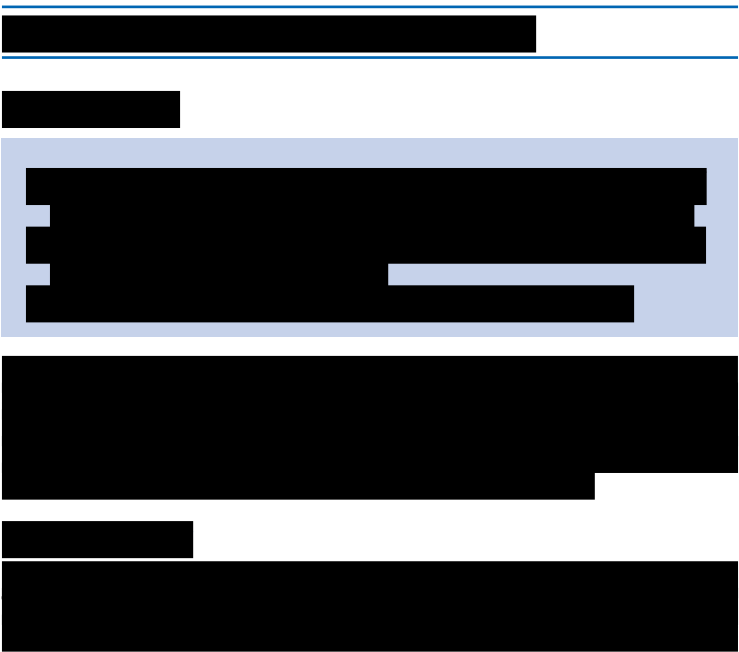
The open nasal/septal repair refers to an aggressive approach to the acute management of complicated injuries using existing lacerations or external and intranasal incisions. With this approach the surgeon can reduce, graft, and fixate fractured anatomic components under direct visualization.

Formal Septorhinoplasty

Formal septorhinoplasty after nasal fracture is employed in two general clinical situations. Septorhinoplasty may be used as a delayed primary treatment for late patient presentation or persistent deformity and/or obstruction after a period of fracture observation. It may also be used as a secondary procedure after initial acute management with persistent deformity or obstruction. Precise treatment plan is selected based on anatomic diagnosis of persistent deformities and may include osteotomies and repositioning of the bony pyramid, septoplasty, and grafting.

◆ **Outcome and Follow-Up**

Up to half of untreated nasal fractures require delayed surgical intervention. Immediate closed manipulation reduces the need for delayed intervention in appropriate cases. After nasal bone reduction, externally stabilizing dressings should be applied similarly to standard rhinoplasty care, and the patient should avoid activities that may lead to nasal trauma for 4 to 6 weeks. When there have been significant septal injuries, internal soft Silastic splints may provide stabilization and aid in the prevention of synechiae. Nasal bones may be supported by intranasal supports placed under the nasal bones. Packing is rarely necessary. Patients should be followed until healing is complete with satisfactory functional and aesthetic results.





9.1.5 Midface Fractures

◆ Key Features

- Midface fractures typically result from high-energy blunt trauma, such as motor vehicle accidents and altercations.
- The mass, density, and speed of the striking object will affect the type and the severity of the facial injury.
- Suspect associated ocular, intracranial, and cervical spine injuries.

The midface transmits masticatory forces to the skull base through a series of three paired vertical pillars, or buttresses, of thickened bone: the zygomaticomaxillary (lateral), nasomaxillary (medial), and pterygomaxillary (posterior) buttresses (**Fig. 9.5**). These buttresses are essential for proper facial form and function. The horizontal buttresses also provide facial shape and include the frontal bar and supraorbital rims, the infraorbital rims with nasal bones, and the hard palate. Damage to the midface may involve fractures of any of these buttresses and requires high-force injury.

Proper realignment of the vertical buttresses is critical in establishing premorbid dental occlusion, facial height, and projection. Proper alignment of the orbital rims and maxillary alveolus (palatal fractures) is essential for establishing facial width, while the zygomatic arches play a key role in restoring both facial projection and width.

In 1901, René Le Fort described three predominant types of midface fractures, which he classified as follows (**Fig. 9.6**):

- Le Fort I fractures (horizontal) extend above the dental apices from the zygomaticomaxillary buttress through the nasomaxillary buttress, piriform aperture, and nasal septum.
- Le Fort II fractures (pyramidal) extend through the zygomaticomaxillary buttress, inferior orbital rim and orbital floor, the frontal process of the maxilla, and through the nasofrontal suture.
- Le Fort III fractures (craniofacial disjunctions) follow a fracture pattern extending from the nasofrontal suture along the medial orbital wall, then

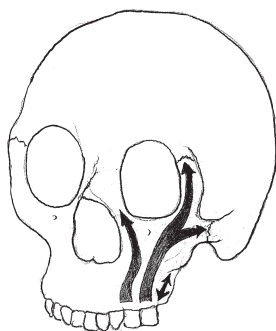


Fig. 9.5 Masticatory forces are transmitted through the zygomaticomaxillary (lateral), nasomaxillary (medial), and pterygomaxillary (posterior) buttresses to the skull base. (Used with permission from Stewart MG, ed. *Head, Face, and Neck Trauma: Comprehensive Management*. New York, NY: Thieme;2005:78.)

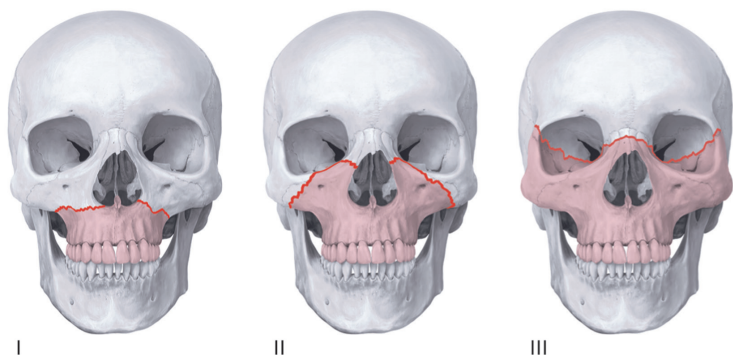


Fig. 9.6 Le Fort classification of midfacial fractures (I, II, III). The framelike construction of the facial skeleton leads to characteristic patterns of fracture lines in the midfacial region. (From THIEME Atlas of Anatomy, Head and Neuroanatomy, ©Thieme 2010. Illustration by Karl Wesker.)

traversing the orbital floor and lateral wall before extending through the zygomaticofrontal suture and zygomatic arch.

◆ Epidemiology

Most midface fractures occur in young males and account for 6 to 25% of facial fractures. Motor vehicle collisions and assaults are the most common mechanisms of injury. In older patients, motor vehicle collisions and falls are the typical cause.

◆ Clinical

Signs and Symptoms

Patients typically present with a history of high-force traumatic facial injury and may complain of pain, malocclusion, trismus, hypoesthesia, or diplopia. Examination often reveals periorbital ecchymosis, facial and oral mucosal edema, facial asymmetry, epistaxis, mobility of the midface, malocclusion

(anterior open-bite deformity), or dental injury. Less commonly, cerebrospinal fluid (CSF) rhinorrhea and changes in visual acuity or restriction of extraocular motion are observed. Airway distress associated with severe injuries and decreased level of consciousness associated with intracranial injury may also be present. Inquiry should be made as to preinjury malocclusion or orthodontic/orthognathic treatment if possible.

Differential Diagnosis

Malocclusion is also seen in isolated dentoalveolar fractures and mandible fractures. Adjacent facial structures must also be evaluated for traumatic involvement (mandible, zygomaticomaxillary, naso-orbito-ethmoid [NOE], frontal sinus, and skull base fractures).

◆ Evaluation

Physical Exam

As with all trauma patients, the initial evaluation should include a detailed systemic examination using the advanced trauma life support protocol, and then a complete head and neck examination is performed. Airway compromise secondary to mucosal trauma and edema or profuse hemorrhage is possible, requiring securing of the airway. The patient's facial region should be thoroughly inspected and palpated for any bony step-offs. Midface mobility is assessed by applying traction to the maxillary central incisors and alveolus with one hand while stabilizing the forehead with the other. All teeth should be accounted for, and if any are fractured or missing, a chest radiograph should be performed to make sure the missing teeth did not enter the airway. A complete cranial nerve (CN) examination is critical. In the midface, injury to the maxillary division of the trigeminal nerve (CN V₂) is common, and facial nerve (CN VII) injury is also possible. Any deficits are documented prior to surgery. Visual acuity and extraocular motion are evaluated and ophthalmologic consultation obtained if these are abnormal or if ocular injury is suspected. Intranasal examination should assess for septal hematoma, septal perforation, and CSF rhinorrhea.

Imaging

All patients with suspected midfacial fractures should undergo a fine-cut multiplanar computed tomography (CT) scan with 3D reconstruction if available. High-quality reformatted coronal images are acceptable when cervical immobilization prevents direct coronal imaging. Many surgeons now routinely obtain CT images immediately postoperatively to confirm anatomic reduction and plating of the fractures. This is extremely useful in panfacial or comminuted fractures.

Labs

No specific laboratory tests are required for this patient population; however, appropriate screening tests (complete blood count [CBC], chemistry, coagulation studies, drug screening) are often obtained during the initial trauma evaluation and may identify conditions warranting treatment.

Other Tests

When CSF rhinorrhea is suspected, β -2 transferrin testing should be performed on the nasal fluid to confirm the diagnosis.

◆ Treatment Options

The goal of treatment is restoration of preinjury function and facial aesthetics. Treatment within the first 7 to 14 days of injury allows tissue edema to subside and lessens the likelihood of aesthetic and functional deficits, which are challenging to correct after delay.

Medical

Midface fractures that are nondisplaced, stable, and accompanied by normal occlusion can be observed, but the mainstay of treatment for all other midface fractures is surgical.

Surgical

Historically, midfacial fractures were managed with a prolonged course of maxillomandibular fixation (MMF) and/or suspension wires. Continued mobility of the fracture lines with this type of treatment led to a high incidence of residual bony and soft tissue deformity. As such, the standard treatment for displaced, mobile, or comminuted midface fractures is now open reduction and internal fixation (ORIF).

Operative treatment begins with exposure of all fracture lines using surgical approaches that may include gingivolabial (maxillary face and buttresses), transconjunctival or subciliary (orbital rim and floor), upper blepharoplasty (zygomaticofrontal and zygomaticosphenoid sutures), and coronal (zygomatic arches, frontal bone, and NOE area) incisions. The maxilla is then disimpacted and proper occlusion reestablished with MMF.

Reduction of all fractures is then performed along with internal fixation with titanium miniplates along the medial and lateral buttresses and along the inferior orbital rims, zygomaticofrontal sutures, zygomatic arches, and glabellar region as indicated. Consideration should be given to primary bone grafting when interfragmentary gaps > 5 mm are present. Miniplates (typically 1.5–2.0 mm) are used for medial and lateral buttresses. Microplates (1.0–1.3 mm) are used on the inferior orbital rim, and the zygomaticofrontal suture is treated with 1.3- to 1.5-mm plates. Although rigid or semirigid plating has greatly improved our ability to treat these injuries, the surgeon must be meticulous in achieving correct anatomic reduction and use exacting technique when adapting the plate to the bone to avoid “fixing” the patient in the wrong position. Most patients are released from MMF at the conclusion of surgery, although arch bars may be left in place for guiding elastics (occasionally useful with severe edema).

◆ Complications

Complications may include injury to sensory or motor nerves. It is essential to document nerve function prior to surgery. Most patients with sensory change will see improvement over the next 3 to 6 months. Temporary

paresis of the temporal branch of the facial nerve may be seen if prolonged retraction during a coronal approach is required, but with careful exposure and surgical technique, permanent injury is rare.

Malocclusion should be noted postoperatively and surgical technique reviewed. CT may be helpful in determining the cause of the malocclusion, and if fracture malalignment is noted, reexploration should be considered, with revision reduction and plating prior to fracture union. Guiding elastics may assist in very minor dental malocclusions but cannot correct grossly malaligned fractures. Persistent facial deformity after bony union may require complex revision surgery with osteotomies or custom implant placement.

Lower eyelid malposition is largely avoided with gentle tissue handling and meticulous surgical technique. Ectropion is more common with transcutaneous than transconjunctival approaches, though entropion may be observed after transconjunctival approaches (rare). Massage is useful in the postoperative period if lower eyelid retraction is noted. More severe malpositions may require surgical intervention.

Other incisional problems include dehiscence, local infection, hypertrophy, and alopecia and are treated as indicated. Hardware exposure, infection, and loosening may require surgical removal after bony union is achieved. Hardware may be palpable. Nonunion and malunion are rare in midface fractures.

◆ Outcome and Follow-Up

The patient is admitted after surgery and monitored for airway status, visual change, hemorrhage, and pain control. Intermittent ice application is useful in decreasing edema, as is elevation of the head of the bed. The patient is instructed in oral hygiene consisting of frequent mouth rinses and gentle brushing, especially after meals. Diet should consist of soft foods for approximately 6 weeks, and then a normal diet may be resumed. Cutaneous sutures are removed ~ 7 days after surgery. Intraoral incisions are closed with resorbable sutures and do not require removal.

The patient should be followed closely for the first several weeks, with particular attention given to occlusal status and eyelid position. Long-term follow-up should also be scheduled, but it can sometimes be challenging in this patient population.

9.1.6 Mandible Fractures

◆ Key Features

- Pain and malocclusion are common findings in mandible fractures.
- Dentoalveolar trauma is often associated.
- Favorable fractures may be treated with closed reduction.
- Open reduction with internal fixation (ORIF) is often required for complex, comminuted, or multiple fractures.

Mandible fractures are common after facial trauma. Diagnosis can often be made by clinical exam, although confirmatory imaging evaluation is required. Fractures often traverse the alveolus, creating intraoral communication with the fracture site, which leads to contamination of fractures by oral flora. Antibiotic use is controversial. Repair is aimed at restoring premorbid occlusion, reestablishing anatomic alignment of the bone fragments, and ensuring healing with minimal morbidity. Mandible fractures are classified as favorable or unfavorable for healing (**Fig. 9.7**).

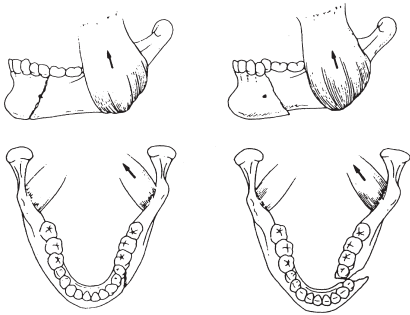


Fig. 9.7 Favorable and unfavorable mandible fractures. (Used with permission from Papell ID, ed. *Facial Plastic and Reconstructive Surgery*. 3rd ed. New York, NY: Thieme;2009:1002.)

◆ Epidemiology

Most mandible fractures are the result of interpersonal trauma or motor vehicle accidents; they typically occur in the third and fourth decades of life. In the elderly, falls become a more common cause, and in younger children sports activities and motor vehicle accidents are most prevalent.

Most studies indicate that the condylar/subcondylar complex is most frequently fractured, followed by the symphyseal region from canine to canine (**Fig. 9.8**). The mandibular angle region is next, followed by the body region. Fractures of the vertical ramus (excluding subcondylar fractures) occur less frequently, and coronoid process fractures are distinctly uncommon. There is some controversy regarding location frequency.

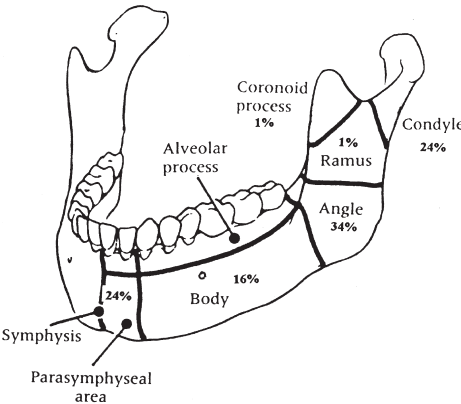


Fig. 9.8 Anatomy of the mandible, with relative incidence of fracture sites. (Used with permission from Papell ID, ed. *Facial Plastic and Reconstructive Surgery*. 3rd ed. New York, NY: Thieme;2009:1001.)

◆ Clinical

Signs

Malocclusion, with or without trismus, may be present. Edema and ecchymosis may be seen both intraorally and extraorally. Loose or fractured teeth should be identified, and attempts should be made to account for missing teeth (intruded teeth may occasionally be mistaken for avulsions). Intraoral lacerations are common at fracture sites. Paresthesia over the chin indicates injury to one or both inferior alveolar nerves.

Symptoms

Most patients with mandible fractures present because of pain or malocclusion, both of which tend to interfere with eating. It is important to establish what the patient's premorbid occlusion was, if possible. Other complaints include oral bleeding, facial swelling, loose teeth, jaw mobility, chin numbness, and trismus.

Differential Diagnosis

If the patient is (or was) unconscious, the nature of the trauma may be difficult to determine. Swelling and muscle injury may masquerade as a fracture. A patient with a serious dental infection, with or without a history of trauma, may have clinical findings of pain, trismus, and malocclusion, which may make it difficult to determine whether a fracture is present. Dislocation of the temporomandibular joints may present with similar symptoms. Other maxillofacial injuries should also be ruled out.

◆ Evaluation

Physical Exam

An initial trauma survey and complete head and neck exam should be performed. If any concern for airway compromise, the airway should be secured. Examine for swelling, tenderness, and ecchymosis. Warmth and fever would be signs of infection, usually seen when presentation is delayed. Examination of the mandible itself might reveal mobility of fragments (to bimanual palpation), trismus, and malocclusion. The occlusion class is based on the relationship of the retrobuccal cusp of the upper first molar to the buccal groove of the lower first molar:

- Class I occlusion (normal): The mesiobuccal cusp of the maxillary first molar occludes exactly with the mandibular first molar buccal groove.
- Class II occlusion: The mesiobuccal cusp is mesial or anterior to the mandibular first molar buccal groove.
- Class III occlusion: The mesiobuccal cusp is distal to the buccal groove.

See **Fig. 9.9**. Also look for loose, cracked, avulsed, and/or impacted teeth. A cranial nerve (CN) exam may show mandibular nerve (CN V₃) hypoesthesia on one or both sides.

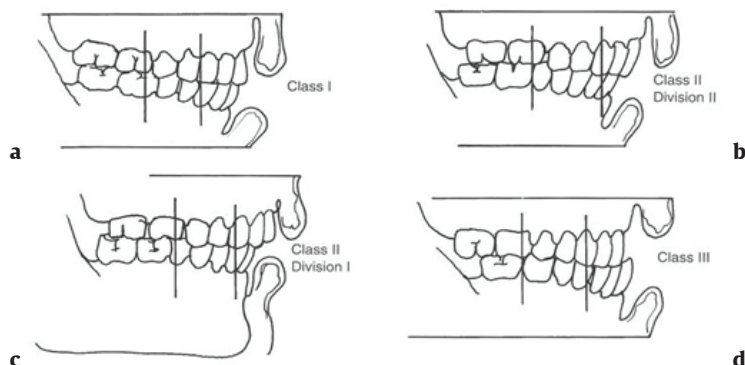


Fig. 9.9 Variants of adult occlusion. **(a)** Class I occlusion or ideal occlusion. **(b,c)** Variants of Class II occlusion. **(d)** Class III occlusion. (Used with permission from Papell ID, ed. *Facial Plastic and Reconstructive Surgery*. 4th ed. New York, NY: Thieme;2016:836.)

Imaging

Plain films are rarely used. A panoramic oral X-ray (Panorex, Dental Imaging Technologies Corp., Hatfield, PA) or fine-cut multiplanar computed tomography (CT) scan (maxillofacial, mandible, or Panorex) should be obtained. Some fractures may be difficult to visualize on Panorex, and in most settings a CT scan is easily acquired and becoming the study of choice.

Labs

No specific labs are required; however, standard preoperative labs may be obtained prior to surgical intervention (complete blood count [CBC], chemistry, coagulation studies, toxicology screen).

Pathology

Pathology studies are generally not applicable, unless there is a concern about a pathologic fracture in a patient with another underlying disease, or in a case of delayed management or complications of prior management, in which there may be concern for posttraumatic mandibular osteitis or osteomyelitis.

◆ Treatment Options

Medical

Nondisplaced, nonmobile favorable fractures without malocclusion can be managed nonsurgically. The patient is maintained on a mechanical soft diet. The patient should be seen after 1 to 2 weeks to ensure patient compliance and that the fracture is healing uneventfully. Any change in occlusion status will require additional treatment.

Subcondylar fractures with minimal displacement and without malocclusion may be managed with physiotherapy and exercises.

For both of the situations just mentioned, mouth-opening exercises are important to prevent trismus.

Relevant Pharmacology

Antibiotic prophylaxis is controversial. Despite a lack of good evidence to support their use, most patients are treated at some point in their care with an antimicrobial agent (preoperatively, perioperatively, or postoperatively). Actively infected wounds should be treated with an antibiotic that covers common oral flora with culture-directed antibiotics if possible. Patients with osteomyelitis may require débridement and prolonged intravenous antibiotics.

Surgical

There are three basic surgical approaches to mandible fractures: (1) placement of oral appliances and application of rigid maxillomandibular fixation (MMF; also called closed reduction, rigid fixation); (2) placement of oral appliances and the use of training elastics (nonrigid MMF; also called non-rigid closed reduction); and (3) open reduction of the fractures, generally used with rigid fixation of the fragments (ORIF).

The most common appliances used for MMF in the United States today are Erich arch bars. These are generally fixed to the teeth with wires and are equipped with small hubs to allow the placement of wires or elastic bands for rigid and nonrigid MMF, respectively. In the past couple of decades, multiple other techniques for MMF have been developed, including intermaxillary screws and multiple hybrid MMF systems that use screws rather than wire fixation of an arch bar. MMF is generally reserved as a solitary treatment option for uncomplicated, isolated, or favorable fractures as well as unilateral subcondylar fractures. MMF is often used intraoperatively to obtain and maintain occlusion prior to rigid fixation.

ORIF may be performed via either intraoral or extraoral approach. "Load-sharing" repairs are generally used when the bone being repaired is solid enough to provide a buttress, so that fixation can be placed in a biomechanically advantaged fashion that takes advantage of the naturally occurring forces of muscle function and mastication. This can be accomplished using the "Champy" technique, with miniplates placed along ideal lines of osteosynthesis with monocortical screws to avoid injury to tooth roots and the inferior alveolar nerve. This technique is commonly used in angle fractures. Two plates or two lag screws are generally used in the symphyseal and parasymphyseal regions, a single plate is commonly used along the mandibular body, and one or two plates are used for angle fractures. Compression plates can be used along the symphysis and body (not at the angle) as well, but this requires that bicortical screws be placed along the inferior border, so tension band plates or arch bars must be applied to avoid distraction of the alveolar portion of the fracture.

“Load-bearing” repairs are used when there is inadequate bone to form a buttress and share in the load. This requires the placement of longer, stronger reconstruction plates fixed with bicortical screws along the inferior border of the mandible. At least three and preferably four screws should be placed on either side of the fracture. Load-bearing reconstruction plate repairs are indicated to span areas of mandibular deficiency, such as defects, areas of comminution, atrophic mandibles (edentulous patients), and areas involved with infection (or previous nonunion). The reconstruction plate is also a fallback technique for any mandible fracture, particularly in the angle region after loss of an impacted third molar. Fractures of the condylar neck should be opened if there is significant foreshortening of the ramus of the mandible or persistent malocclusion. The endoscopic approach allows a mostly transoral repair of selected subcondylar fractures. Open reduction of condylar head fractures remains quite controversial.

Finally, external fixation with placement of a percutaneous external fixator device may be required in severe injuries, significant comminution, or with nonunion/hardware infection.

◆ Complications

A rigidly fixed malreduction should be reoperated, unless there are extenuating circumstances, because MMF cannot repair it. Wound infections should be drained and managed expectantly. Loose hardware must be removed. Failure of fixation requires reoperation, and if infection has developed, a stronger, load-bearing repair will be necessary. Nerve injuries (motor and sensory) should be documented.

◆ Outcome and Follow-Up

Routine wound care is indicated. Oral hygiene must be maintained, and antiseptic oral rinses are commonly used several times daily and after meals. A liquid diet is preferred initially, and this is advanced to a mechanical soft diet as tolerated. The use of postoperative imaging to ensure satisfactory reduction of fractures is controversial. If there is any concern for persistent malocclusion or hardware complication, imaging should be obtained. If satisfactory rigid fixation has been accomplished, MMF will not be necessary. After 1 week, trismus should be treated with mouth-opening exercises. Patients treated with rigid MMF are generally sent to the floor and discharged with wire cutters, in case they need to urgently release the MMF.

Patients should be followed closely for the first 6 weeks. At this point, most fractures are stable enough to allow removal of the arch bars. After this, patients should be followed until normal function is ensured.

9.2 Facial Paralysis, Facial Reanimation, and Eye Care

◆ Key Features

- No single modality is universally appropriate for all afflictions of facial nerve function.
- Etiology of the paralysis, oncologic status, type of injury, and location of injury all contribute to the selection of the most appropriate treatment methods.
- Eye protection is critical.
- Close patient follow-up and counseling are necessary.

An extensive list of possible etiologic factors for facial paralysis exists. This may be narrowed based on clinical history and exam to direct further work-up. Treatment options may include medical therapies, rehabilitation, static procedures, dynamic procedures, and reanimation. Close follow-up and the setting of realistic expectations are important.

◆ Epidemiology

Facial paralysis may affect individuals of any age but is more common in the fifth to sixth decade. Females are more commonly affected than males.

◆ Clinical

Signs

Facial weakness may occur in select or all branches of the facial nerve. Decreased blink reflex, incomplete eye closure, or lower eyelid rounding may be observed. Deviation of nasal base and philtrum, loss of melolabial fold definition, or a drooped oral commissure are common. Patients may have periauricular vesicles, a body rash, or other neurologic deficits.

Symptoms

A detailed history is the key to identifying the etiologic cause of the facial weakness. A history of preceding illnesses, surgery, or trauma should be elucidated. Travel patterns, particularly in locations endemic with Lyme disease, or a tick bite may be reported. Facial weakness may be described as rapid onset (< 72 hours), delayed onset, progressive, fluctuating, unilateral, or bilateral. It may involve the entire hemiface or select branches. Associated pain or mass raises a concern for malignancy. A full assessment of other neurologic symptoms and a general review of systems should be undertaken.

Differential Diagnosis

There are a myriad of potential causes of facial paralysis. Bell's palsy (idiopathic) is the most common cause and is generally self-limited, with

the majority of patients showing complete resolution (see **Chapter 3.1.3**). Other common causes are viral reactivation (Bell's palsy, Ramsay Hunt's syndrome), infection (Lyme disease, otitis media), or injury. Facial paralysis due to nerve injury may be iatrogenic or traumatic.

Intracranial nerve injuries most commonly occur during resection of vestibular schwannoma or other cerebellopontine angle (CPA) tumors. The incidence of facial nerve injury following CPA tumor surgery is reported to be 2.3%. Intratemporal facial nerve injury is usually encountered in patients following external head trauma with skull base fractures or iatrogenic injury during or following otologic surgery. Most temporal bone fractures result from motor vehicle accidents and violent encounters. Seven to 10% of these fractures result in facial nerve dysfunction, with paralysis more common in transverse fractures. Extratemporal injury to the facial nerve may occur during parotid surgery, temporomandibular joint procedures, or facelift procedures or following traumatic lacerations of the face. Patients at higher risk for facial nerve injury during parotid surgery include children and those undergoing a total parotidectomy. The differential may be narrowed by history, exam, and studies as indicated.

◆ Evaluation

Physical Exam

A general neurologic evaluation, complete head and neck exam, and a detailed cranial nerve exam should be performed, as findings may help narrow the differential diagnosis. At a minimum, the degree of facial impairment should be graded by the House–Brackmann Facial Nerve Grading System (**Table 3.3** in **Chapter 3.1.3**), although more detailed and updated grading systems now exist (**Table 9.2**; **Fig. 9.11**).

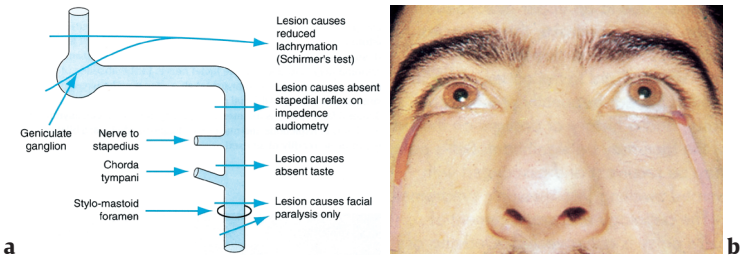


Fig. 9.11 Tests of facial nerve involvement. The level of involvement of the facial nerve in facial palsy can be determined by: **(a)** taste (electrogustometry—if taste is absent or impaired, then the lesion is proximal to the chorda tympani); stapedial reflex (impedance audiometry); or **(b)** lacrimation (Schirmer test litmus paper is placed under the lower lid; if the facial nerve lesion is proximal to or involves the geniculate ganglion, the tears are reduced). These tests are reliable in traumatic section of the facial nerve to detect the level of injury, but in Bell's palsy, these tests are of little value. (Used with permission from Bull TR, Almeyda JS. *Color Atlas of ENT Diagnosis*. 5th ed. New York, NY: Thieme;2010:103.)

Table 9.2 The Sunderland classification of nerve injury

	Type of injury and functional consequence	Structures involved	Management	Prognosis
Class I	Neuropraxia/ compression resulting in temporary dysfunction of Na ⁺ channels, preventing transmission of nerve impulses	Myelin sheath	Watchful waiting ± steroids	Likely full recovery within weeks to months
Class II	Typically a crush or displacement of bony fragments resulting in axonotmesis, with preservation of individual endoneurial channels, resulting in Wallerian degeneration	Myelin, axons	Watchful waiting ± steroids	Likely full recovery over months
Class III	Laceration/ischemic injury resulting in neurotmesis and Wallerian degeneration	Myelin, axon, endoneurium	Surgical repair using suture or fibrin glue	Full recovery unlikely
Class IV	Laceration/ischemic injury resulting in neurotmesis and Wallerian degeneration with damage extended to fascicles, but preservation of the epineurial sheath	Myelin, axon, endoneurium, perineurium	Surgical repair using suture or fibrin glue	Full recovery unlikely
Class V	Laceration/ischemic injury resulting in complete discontinuity of proximal neural elements from distal elements, causing neurotmesis and Wallerian degeneration	Myelin, axon, endoneurium, perineurium, epineurium	Surgical repair using suture or fibrin glue	Full recovery unlikely

Data from Sunderland S. *Nerve Injuries and Their Repair: A Critical Appraisal*. Edinburgh/New York, NY: Churchill Livingstone;1991.

Imaging

Imaging is not routine for all patients, but computed tomography (CT) or magnetic resonance imaging (MRI) may be appropriate, based on history and exam. Facial paralysis in association with other neurologic deficits may warrant imaging to evaluate for stroke or intracranial tumor. Recurrent facial paralysis warrants imaging, typically an MRI scan of the internal auditory canal. Facial paralysis with facial pain or a mass warrants a CT scan or MRI of the face/neck to evaluate for malignancy. Other imaging tests will be performed when clinically indicated.

Labs

Lab studies are not routinely performed on all patients at initial presentation. Specific labs may be ordered based on the clinical scenario and include a complete blood count (CBC), C-reactive protein [CRP], autoimmune labs (antineutrophil cytoplasmic antibody [ANCA], rheumatoid factor [RF], anti-nuclear antibody [ANA]), tests for syphilis, Lyme titers, glucose, and human immunodeficiency virus (HIV).

Other Tests

The use of electrodiagnostic testing (typically electroneurography [ENoG]) is used in select situations for prognosis and occasionally treatment planning. It is helpful > 72 hours after onset of paralysis and within two weeks. If an ENoG shows > 90% degeneration of the facial nerve, a confirmatory electromyogram is typically then performed. If this shows no motor unit potentials, a facial nerve decompression may be considered.

◆ Treatment Options

Medical

If a specific clinical diagnosis is not suggested based on history or exam, most authors recommend an initial treatment with a high-dose steroid taper and antiviral therapy, although the latter is controversial. If an underlying disease is identified based on additional studies or findings, then treatment of the underlying disorder is appropriate (doxycycline for Lyme disease, antiretrovirals for HIV, immune modulators for autoimmune disease).

Surgical

The options (in order of preference) for restoration of function following total unilateral facial paralysis are as follows:

1. Spontaneous facial nerve regeneration (observation)
2. Facial nerve neurotization (facial nerve anastomosis)
3. Interpositional graft (cable graft)
4. Nerve crossovers (anastomosis to other motor nerves)
5. Muscle transfer
6. Static procedures

Dynamic Procedures/Reanimation

● Facial Nerve Neurorrhaphy

If the nerve has been completely disrupted, then direct neurorrhaphy is the most effective way to reanimate the paralyzed face. The interrupted neural pathway can be reestablished either by direct anastomosis or by inserting a graft between the disrupted segments. Some of the key points in nerve repair are early identification, evaluation of nerve condition, and tension-free anastomosis. The best time to perform surgery is within the first 72 hours, before degeneration has occurred and while the distal nerve can still be stimulated, although success with exploration and repair has been noted up to several months after injury. It may be necessary to reroute the nerve within the temporal bone or to gain extra length by releasing the nerve. Factors that influence the success of repair include tension, the character of the wound, the presence of scar tissue, and time lag to repair.

The surgical suturing technique for nerve repair requires magnification, either with loupes or a surgical microscope. The nerve endings should be freshened with a new scalpel blade. At this point, axoplasm may be seen oozing from the proximal stump. Number 8-0 to 10-0 nylon sutures with a 75- or 100-micron needle should be used.

If possible, three or four simple sutures should be placed about the circumference of the epineural layers to achieve adequate union (**Fig. 9.12**).

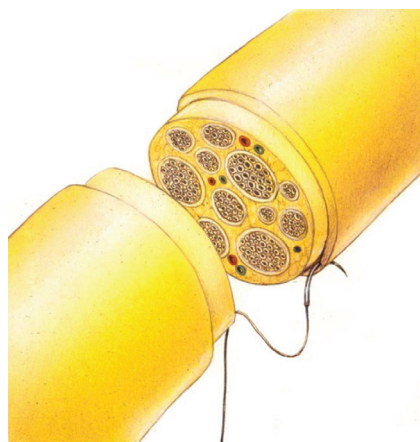


Fig. 9.12 Epineural nerve repair. (Used with permission from Papel ID, ed. *Facial Plastic and Reconstructive Surgery 4th Edition*. New York, NY: Thieme;2016:738.)

● Interpositional Graft

In cases where patients have undergone prior surgery or have had part of their facial nerve sacrificed or avulsed as a result of severe trauma, direct nerve repair is impossible and interposition of a nerve graft is required. This technique is reserved for cases in which direct nerve repair would result in excess tension or when there is loss of nerve tissue.

The great auricular nerve is the most commonly used donor nerve, especially when the nerve graft required is small. Advantages include its proximity to the operative field and ease of exposure (**Fig. 9.13**). If more nerve tissue is required, then the sural nerve may be harvested.

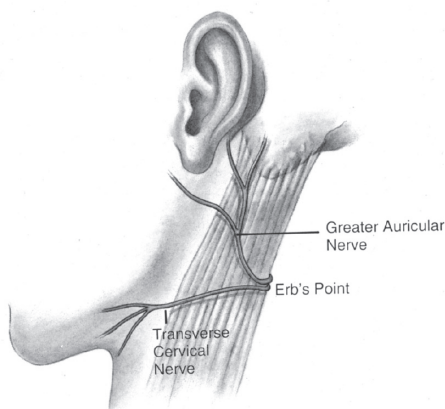


Fig. 9.13 Greater auricular and transverse cervical nerve grafts. (Used with permission from Burgess LPA, Goode RL. *Reanimation of the Paralyzed Face*. New York, NY: Thieme;1994:18.)

● Nerve Crossovers

This technique is used when direct suturing or cable grafting is not feasible, as is often the case after removal or obliteration of proximal or intratemporal portions of the facial nerve. It is particularly useful to treat facial paralysis resulting from intracranial or intratemporal disorders or surgery. These techniques are relatively simple and require one suture line. They provide a powerful source for reinnervation, although the results are not always consistent or predictable. Nerve crossovers most commonly utilize the hypoglossal nerve, the accessory nerve, or, more recently, the masseteric nerve. As the masseteric nerve is within the operative field, is easy to identify, and produces less morbidity than other nerve sources, it is becoming the donor nerve of choice.

Another technique is the “babysitter” graft. In this technique, the hypoglossal nerve and facial nerve are anastomosed with the interposition of a free nerve graft, end-to-end to the peripheral facial nerve stump and end-to-side to the hypoglossal nerve. This technique is typically used in candidates for cross-facial nerve grafting.

Cross-facial nerve grafting is a technique that uses the contralateral normal facial nerve to innervate certain facial muscles on the paralyzed side. This technique should be considered an alternative to hypoglossal or accessory nerve grafting. It should not be performed as long as spontaneous regeneration is still possible or in cases in which direct or cable grafting of the facial nerve is possible.

● Muscle Transfer Techniques

Muscle transfer techniques are used when neural techniques are unsuitable. Patients with longstanding facial paralysis (> 2 years) are unlikely to benefit from any of the reanimation procedures just discussed. Severe fibrosis occurs in the distal neuromuscular unit, along with atrophy of the facial musculature, making reinnervation unlikely.

○ Regional Muscle Transfer

Muscle transfer techniques entail transplanting a new neuromuscular unit into a region of the paralyzed face. This may be done in conjunction with a nerve graft or a crossover implanted in the transferred muscle. The two basic techniques to accomplish this are regional muscle transposition and a free-muscle transfer.

Regional muscle transfer is usually used to reanimate the lower third of the paralyzed face. The new neuromuscular unit is composed of the transposed muscle with its original nerve supply. Muscles available for these procedures include the masseter, temporalis, and digastric. It should be remembered that in all muscle transfer procedures, overcorrection is desired.

○ Temporalis Tendon/Muscle Transfer

In an attempt to enable some dynamic control of the oral commissure and smile symmetry, a temporalis muscle or tendon transfer may be performed. The temporalis muscle may be used to elevate the corner of the mouth. It classically was performed by exposing the insertion in the temple, folding a segment of the muscle over the zygomatic arch, and securing the muscle to the oral commissure. This technique is losing favor because of the secondary deformity of temporal hollowing and a visible bulge over the zygomatic arch. If used, most providers now favor the temporalis tendon transfer, which incorporates a transoral or transcutaneous incision followed by a coronoidectomy and detachment of the temporalis tendon. The tendon is then secured to the oral commissure musculature. This technique is also used less frequently as reinnervation techniques have gained popularity.

○ Free-Muscle Transfer

The ideal situation for use of an innervated vascularized muscle free flap for facial reanimation is the large defect seen following a radical parotidectomy. This flap would be used for both reanimation and soft tissue defect reconstruction. The free-muscle flap may also be used in cases of long-term paralysis in which considerable muscle atrophy or soft tissue contracture has taken place. The most commonly used free flaps for facial reanimation are the gracilis muscle, the inferior rectus abdominis muscle, and the latissimus dorsi muscle.

Static Procedures

Many procedures do not restore dynamic function of the face or reinnervate the face but allow static improvement in the appearance and function. Static treatments may be used in conjunction with dynamic or reinnervation procedures.

- *Tarsorrhaphy*

Tarsorrhaphy is an effective method of eye protection in patients with facial nerve paralysis and mild lagophthalmos. Functional and cosmetic drawbacks are associated with this type of procedure. A central tarsorrhaphy completely impairs vision and is not cosmetically acceptable as a permanent procedure.

- *Wedge Resection and Canthoplasty*

Wedge resection of the lower lid with canthoplasty is an effective and relatively simple procedure. It is particularly effective when lower lid laxity is mild. In cases with severe lid laxity or ectropion, a canthoplasty procedure is recommended.

- *Gold or Platinum Weights*

The gold or platinum weight implant is a very simple procedure that offers consistently satisfactory results. A gold weight lid load can be placed under local anesthesia and takes advantage of the relaxation of the levator that occurs with attempted eye closure and gravity. The weight in the upper eyelid passively closes the lid. Gold has been the material of choice for lid weighting because of its high density, relative inertness, and color, which blends with most skin tones. In recent years, platinum has become more frequently used, as the same density weight is available in a lower profile form and may have lower levels of complication.

Complications of weight insertion include extrusion, position shift, infection of the graft, contact allergy to gold, and overclosure of the eyelid.

- *Palpebral Spring Implant*

In cases in which there is poor levator action, a palpebral spring implant may prove more efficacious. However, this technique is technically more difficult and has a higher extrusion rate. The palpebral spring is basically a piece of wire and is therefore subject to possible complications, such as breakage, wearing out, migration, extrusion, and infection.

- *Lower Eyelid Shortening*

Lower eyelid weakness, rounding, and ectropion cause epiphora, worsening asymmetry, and increased scleral exposure. A lower eyelid tightening (canthopexy) or shortening procedure may be performed.

They may be performed under local anesthesia. A lateral canthotomy and inferior cantholysis are performed, and the anterior and posterior lamellae

are denuded off the tarsus in addition to the superior epithelial surface. The bare tarsus may be trimmed to shorten the lower lid; then the lateral canthal attachment is recreated by securing it more superiorly to the Whitnall tubercle (marginal tubercle of the zygomatic bone), with improved lower lid apposition to the globe. The lateral canthal angle must be reestablished with a buried suture from the lateral inferior to lateral superior lid gray line.

Complications include ocular injury, recurrent lower lid rounding and ectropion, and abnormal rounding of the lateral canthal angle.

● *Static Sling*

This procedure allows static elevation of the corner of the mouth, creation of a nasolabial fold, and correction of alar collapse with improved symmetry at rest. Autologous tensor fasciae latae is the material of choice, though many authors report use of acellular dermal matrix, cadaveric tissue, or other implants.

Although variations on this procedure exist, in general two to three strips of suspension material are secured to a neo-nasolabial fold and alar base after elevation of a skin flip, then secured to the deep temporal fascia. Overcorrection is typically used, as recurrence has been noted with time.

Complications include infection, necessitating removal of suspension material, and persistent or worsening asymmetry.

● *Temporalis Sling*

The temporalis muscle may be used in similar fashion as a static sling to elevate the corner of the mouth. It is performed by either exposing the insertion in the temple and folding a segment of the muscle over the zygomatic arch and securing the muscle to the oral commissure. This technique is losing favor because of the secondary deformity of temporal hollowing and a visible bulge over the zygomatic arch.

● *Temporalis Tendon Transfer*

In an attempt to allow some dynamic control of the oral commissure and smile symmetry, a temporalis tendon transfer may be performed. This technique incorporates a transoral or transcutaneous incision followed by a coronoidectomy and detachment of the temporalis tendon. The tendon is then secured to the oral commissure musculature. This technique is also utilized less frequently, as reinnervation techniques are gaining popularity.

● *Ancillary Techniques*

Facial nerve paralysis may be associated with involuntary facial movements after partial or complete recovery or after reinnervation procedures (synkinesis). These signs may be due to aberrant regeneration of the facial nerve. Botulinum toxin (Botox, Allergan, Dublin, Ireland) induces a temporary and reversible neuromuscular blockade and thus is useful in alleviating synkinesis. Highly selective neurectomies and select myectomies (e.g., platysmectomy) might be also be useful.

◆ **Eye Care**

Paralysis of the upper branches of the facial nerve results in disorders of the eyelid and lacrimal function. Sequelae include incomplete closure of the eye with corneal exposure, lower lid ectropion with epiphora, decreased tear production, and loss of the corneal “squeegee” effect. These factors contribute to inadequate corneal protection, which can result in exposure keratitis, corneal ulceration, and blindness. Management of the eye in a patient with facial paralysis begins with supportive care to protect the cornea. This includes mainly moisturizing the eye and preventing exposure. These measures should be adequate in cases that are temporary or partial. Artificial tears are commonly used to keep the eyes moist. Ointments should be supplemented, especially at night before sleep. Closure of the eye can be achieved by carefully taping both the upper and lower lids. A clear humidity chamber provides moisture and protects the eye from trauma and foreign bodies.

◆ **Outcome and Follow-Up**

Facial rehabilitation is critical to optimizing results after reanimation procedures and in preventing and decreasing synkinesis. This ideally involves a therapist with specialized training in facial rehabilitation. Techniques used include facial muscle relaxation, stretching, and biofeedback. The goals of reanimation are facial symmetry, eye closure, oral competence, and voluntary movement. None of the described procedures can completely restore the paralyzed face to its normal function. Some synkinesis and residual weakness may persist, yet significant improvements in both function and appearance can be accomplished if the goals of reconstruction are kept in mind. However, patients require long-term follow-up for monitoring of outcomes, with additional procedures as necessary.

